

NFL OUTCOME PREDICTOR

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DATA AND OBJECTIVE

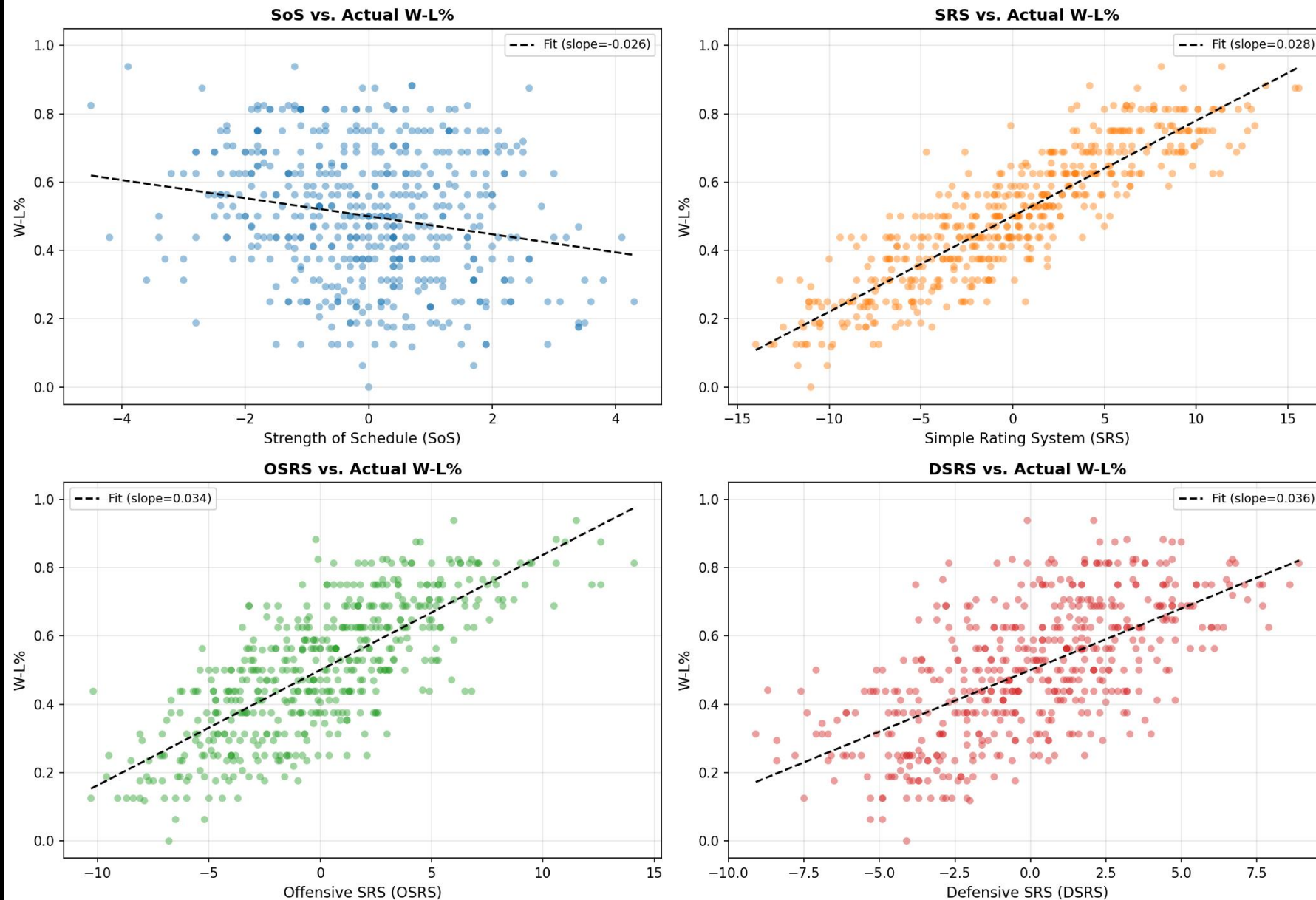
- Pro Football Reference, dataset from: pro-football-reference.com
 - 2010-2025 Regular Seasons
 - **SoS -- Strength of Schedule**
Average quality of opponent as measured by SRS (Simple Rating System)
 - **SRS -- Simple Rating System**
Team quality relative to average (0.0) as measured by SRS (Simple Rating System)
 $SRS = MoV + SoS = OSRS + DSRS$
 - **OSRS -- Offensive SRS**
Team offense quality relative to average (0.0) as measured by SRS
 - **DSRS -- Defensive SRS**
Team defense quality relative to average (0.0) as measured by SRS
- Compare 4 models: Discover which model can most accurately predict a team's **W-L%**
- **60% training, 20% validation, 20% testing. All the same for each model tested.**

EXAMPLE INPUTS AND OUTPUTS

	Tm	SoS	SRS	OSRS	DSRS
0	New England Patriots	2.6	15.4	12.6	2.8
1	New York Jets	2.5	6.5	2.2	4.2
2	Miami Dolphins	4.1	0.3	-2.8	3.1
3	Buffalo Bills	4.3	-4.6	-1.5	-3.1
4	Pittsburgh Steelers	1.3	10.2	2.5	7.7
...	...	...	...	...	...
27	New Orleans Saints	-0.3	-4.9	-5.5	0.6
28	Seattle Seahawks	1.6	12.8	6.2	6.7
29	Los Angeles Rams	2.4	12.5	8.7	3.9
30	San Francisco 49ers	2.1	6.0	3.7	2.3
31	Arizona Cardinals	3.4	-4.4	-1.0	-3.4

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.176	0.058824
1	Atlanta Falcons	0.471	0.470588
2	Baltimore Ravens	0.471	0.529412
3	Buffalo Bills	0.706	0.882353
4	Carolina Panthers	0.471	0.294118
5	Chicago Bears	0.647	0.764706
6	Cincinnati Bengals	0.353	0.235294
7	Cleveland Browns	0.294	0.176471
8	Dallas Cowboys	0.441	0.470588
9	Denver Broncos	0.824	0.882353
10	Detroit Lions	0.529	0.941176

NFL Team Features vs. Actual W-L% (2010-2025)



ORIGINAL DATASET

MODELS USED

- Logistic Regression – Greg Yonan
- Neural Network – Dylan Shaffer
- Random Forest – Simon Liedtke
- Naïve Bayes – Dilen Patel

LOGISTIC REGRESSION

- **Data Input**
 - Team stats are gathered from Pro Football Reference
 - CSV files per season
- **Model Training**
 - Features: PD, SoS, SRS, OSRS, DSRS
 - Target: W-L%
 - Train/test/validation split (60/20/20)
- **Prediction Logic**
 - Predict team strength
 - Compare two teams, and the higher predicted value wins
- **Output**
 - Predicted vs Actual W-L%
 - Saved as CSV

Model saved.

R^2 : 0.8391

MSE: 0.006283

Mean % Error: 16.76%

	Team	Actual W-L%	Predicted W-L%
1	New York Jets	0.176	0.0
2	Tennessee Titans	0.176	0.0
3	Arizona Cardinals	0.176	0.058823529411764705
4	Las Vegas Raiders	0.176	0.058823529411764705
5	Washington Commanders	0.294	0.058823529411764705
6	Cincinnati Bengals	0.353	0.11764705882352941
7	New Orleans Saints	0.353	0.17647058823529413
8	Atlanta Falcons	0.471	0.29411764705882354
9	Cleveland Browns	0.294	0.29411764705882354
10	Dallas Cowboys	0.441	0.29411764705882354
11	Baltimore Ravens	0.471	0.35294117647058826
12	Indianapolis Colts	0.471	0.35294117647058826
13	New York Giants	0.235	0.35294117647058826
14	Tampa Bay Buccaneers	0.471	0.35294117647058826
15	Miami Dolphins	0.412	0.47058823529411764
16	Green Bay Packers	0.559	0.5294117647058824
17	Los Angeles Chargers	0.647	0.5294117647058824
18	Pittsburgh Steelers	0.588	0.5294117647058824
19	Carolina Panthers	0.471	0.5882352941176471
20	Kansas City Chiefs	0.353	0.6470588235294118
21	Minnesota Vikings	0.529	0.6470588235294118
22	San Francisco 49ers	0.706	0.6470588235294118
23	Houston Texans	0.706	0.7058823529411765
24	Chicago Bears	0.647	0.7647058823529411
25	Buffalo Bills	0.706	0.8235294117647058
26	Detroit Lions	0.529	0.8235294117647058
27	Philadelphia Eagles	0.647	0.8235294117647058
28	Jacksonville Jaguars	0.765	0.8823529411764706
29	Los Angeles Rams	0.706	0.8823529411764706
30	Denver Broncos	0.824	0.9411764705882353
31	New England Patriots	0.824	1.0
32	Seattle Seahawks	0.824	1.0

PREDICTED
NFL
STANDINGS

ACTUAL NFL
STANDINGS

	W	L	I	PCT ^
e --  Las Vegas Raiders	3	14	0	.176
e --  New York Jets	3	14	0	.176
e --  Tennessee Titans	3	14	0	.176
e --  Arizona Cardinals	3	14	0	.176
e --  New York Giants	4	13	0	.235
e --  Cleveland Browns	5	12	0	.294
e --  Washington Commanders	5	12	0	.294
e --  Cincinnati Bengals	6	11	0	.353
e --  Kansas City Chiefs	6	11	0	.353
e --  New Orleans Saints	6	11	0	.353
e --  Miami Dolphins	7	10	0	.412
e --  Dallas Cowboys	7	9	1	.441
z --  Carolina Panthers	8	9	0	.471
e --  Indianapolis Colts	8	9	0	.471
e --  Baltimore Ravens	8	9	0	.471
e --  Tampa Bay Buccaneers	8	9	0	.471
e --  Atlanta Falcons	8	9	0	.471
e --  Minnesota Vikings	9	8	0	.529
e --  Detroit Lions	9	8	0	.529
y --  Green Bay Packers	9	7	1	.559
z --  Pittsburgh Steelers	10	7	0	.588
z --  Chicago Bears	11	6	0	.647
z --  Philadelphia Eagles	11	6	0	.647
y --  Los Angeles Chargers	11	6	0	.647
y --  Los Angeles Rams	12	5	0	.706
y --  Houston Texans	12	5	0	.706
y --  Buffalo Bills	12	5	0	.706
y --  San Francisco 49ers	12	5	0	.706
z --  Jacksonville Jaguars	13	4	0	.765
z --  New England Patriots	14	3	0	.824
* --  Denver Broncos	14	3	0	.824
* --  Seattle Seahawks	14	3	0	.824

RESULTS OF LOGISTIC REGRESSION FOR 2025 SEASON

NEURAL NETWORK

- Input -> Two Hidden Layers (64 Neurons, ReLU) -> Output

```
1 def build_model(input_dim):
2     model = keras.Sequential([
3         layers.Dense(64, activation="relu", input_shape=(input_dim,)),
4         layers.Dense(64, activation="relu"),
5         layers.Dense(1) # single output: W-L%
6     ])
7     model.compile(optimizer="rmsprop", loss="mse", metrics=["mae"])
8     return model
```

```
mean = X_train_global.mean(axis=0)
std = X_train_global.std(axis=0)

X_train_norm = (X_train_global - mean) / std
X_val_norm = (X_val_global - mean) / std
```

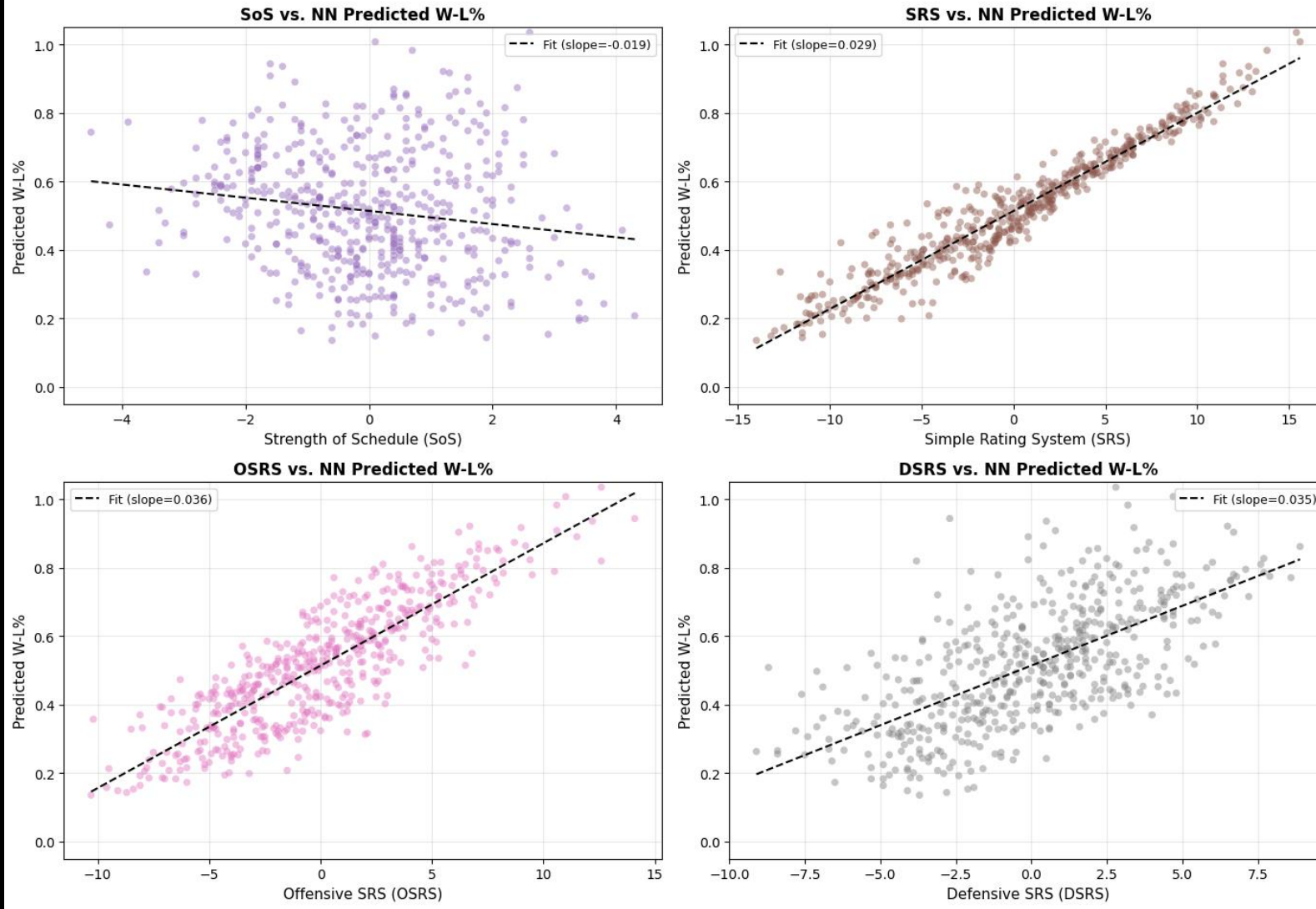
- Trains up to 100 epochs, with early stopping enabled (checking every 10 epochs)

```
Epoch 10 - loss: 0.007902 val_loss: 0.008739 val_mae: 0.076644
Epoch 19: early stopping
Restoring model weights from the end of the best epoch: 9

Done. Best epoch: 9, best val_loss: 0.007407, total epochs: 19
```

```
R2: 0.8255
MSE: 0.006815
Mean % Error: 17.33 %
```


NFL Team Features vs. Neural Network Predicted W-L% (2010-2025)



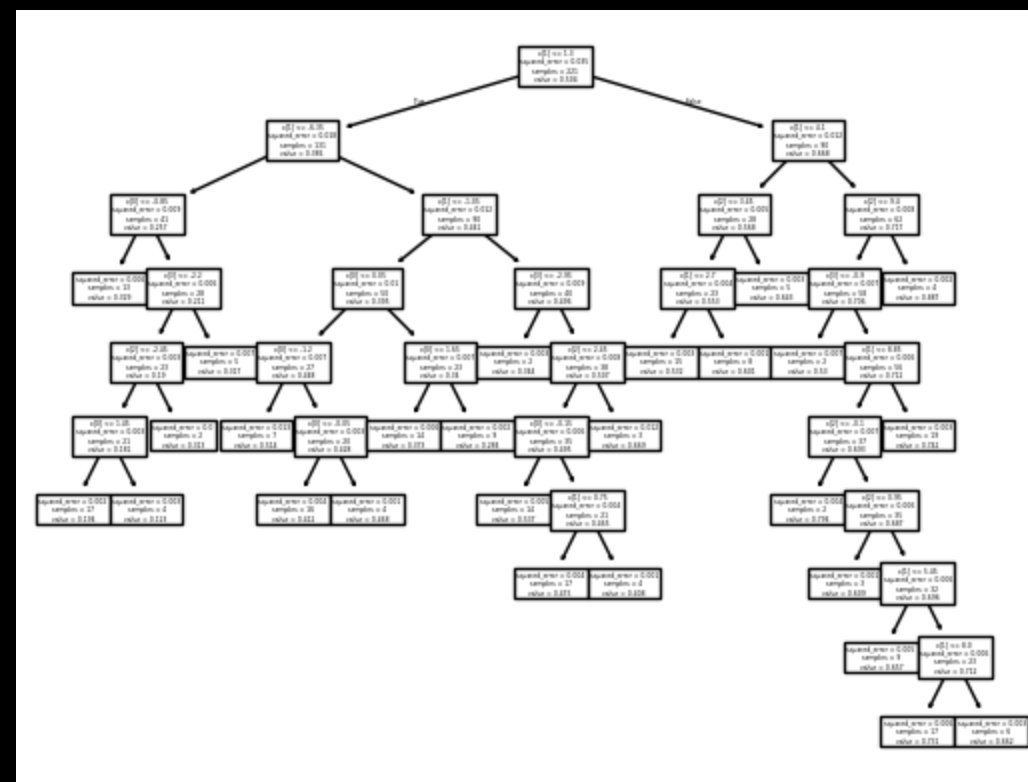
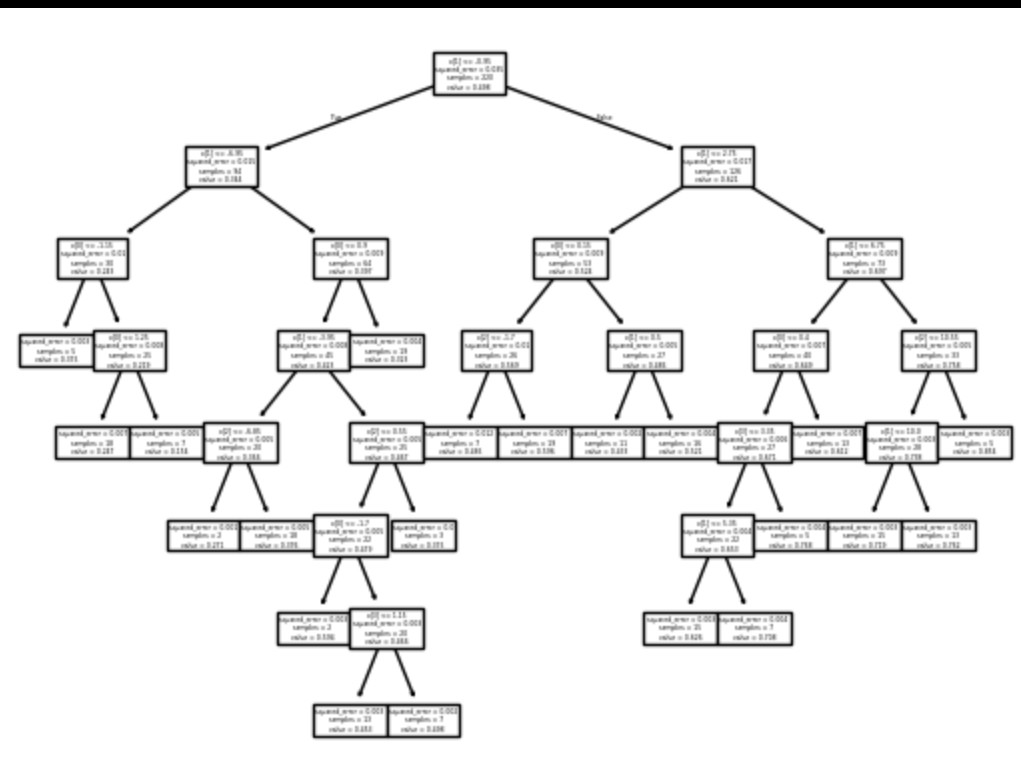
NEURAL NETWORK PREDICTIONS

RANDOM FOREST

	Rand. Forest
MSE	0.007503
R^2	0.8079
Mean Perc. Error	19.17

- Utilized sklearn's random forest regression package, and hyperparameter selection to optimize for the best parameters given the dataset.
- Best hyperparameters: {'n_estimators': 5000, 'min_samples_split': 20, 'min_samples_leaf': 2, 'max_depth': 20, 'bootstrap': True}
- Random forests are a combination of tree predictors such that the prediction strength of each of the trees is averaged over all of the trees, and with an increasing number of trees, the error converges. This gives an in-built ensemble averaged type model and generally captures non-linear relationships well.

EXAMPLES OF TREES IN FOREST



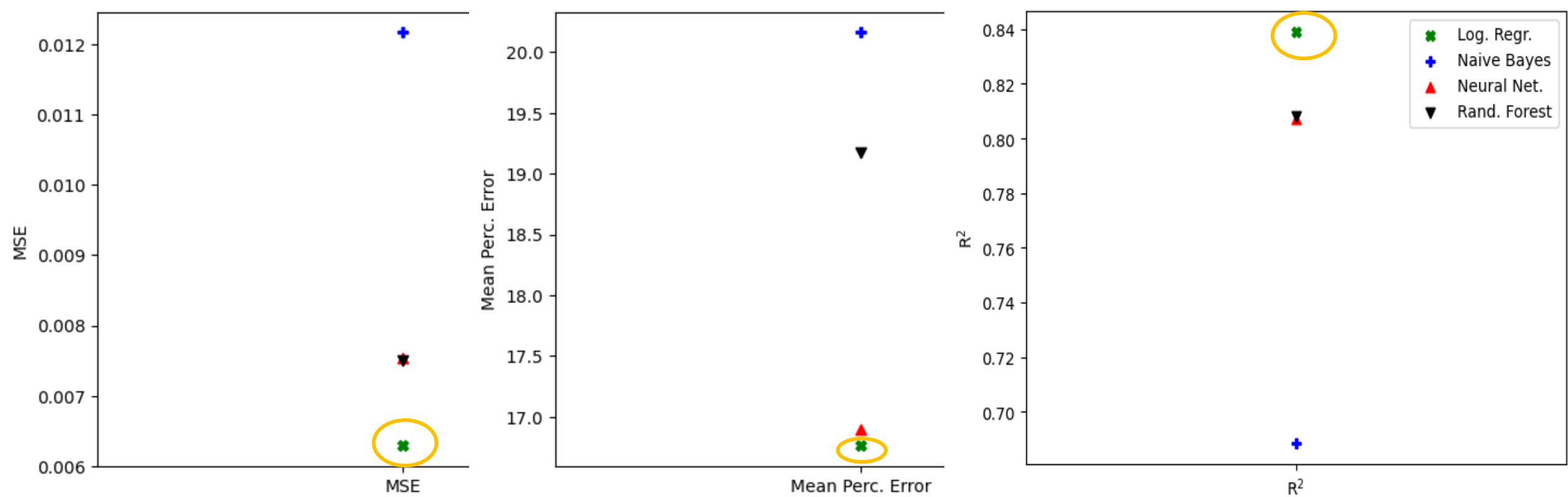
NAÏVE BAYES

- Using the Naive Bayes model, predictions for game outcomes and team season performance in the NFL were made.
- Game and season features included points scored, points allowed, and other statistics.
- In the Naive Bayes model, one uses statistic-related guesswork to determine the most likely game or season outcome based on distributions and priors.
- Performed better than random guessing and OK error metrics, though not as successful as the other models.

	Naive Bayes
MSE	0.012165
R ²	0.6885
Mean Perc. Error	20.16

METRICS AND RESULTS

	Log. Regr.	Naive Bayes	Neural Net.	Rand. Forest	Winner
MSE	0.006289	0.012165	0.007536	0.007503	Log. Regr.
R ²	0.8391	0.6885	0.8070	0.8079	Log. Regr.
Mean Perc. Error	16.76	20.16	16.90	19.17	Log. Regr.



○ = Best

Visualization showing logistic regression outperforms on all metrics.

CONCLUSION

- Logistic Regression performed best on all metrics for prediction of team W/L% given the tested features.
- All models performed quite well in terms of W/L% prediction, clearly outperforming random selection. The lowest R^2 value came from Naïve Bayes at 0.6885 and the best was Logistic Regression at 0.8391.
- Given the dataset size and performance, the Logistic Regression model is near as good a performance as we will find. Moving forward the best way to improve the model and performance would involve finding more features to train on which can predict future W/L% .
- Expanding the model to consider player injuries. This could be done by giving teams overall injury scores based on a numerical value that decides how many key players are injured

THANK YOU